

Let's Get Psychophysiological! A Hands-On Wearables Laboratory Experience with Recording, Processing, and Interpreting Electrophysiology Signals

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Abstract

HCI uses psychological, physiological, and behavioral (i.e. psychophysiological) signals to study user experience and for implicit interaction. Signal acquisition requires understanding of measurement principles that guide interpretation. Psychophysiology considers psychological states and physiological activity to uncover the bidirectional ways that the mind and body intersect. This course features psychophysiological principles, signal acquisition, processing, and interpretation, and hands-on activities. Common signals include heart rate (HR), accelerometry, respiration, brainwaves (EEG), skin conductance (GSR), muscles (EMG), eye-tracking, and sleep (PSG). This year showcases electrocardiography (ECG) and heart rate variability (HRV). Participants will enhance their understanding and ability to use psychophysiology in HCI applications.

CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing systems and tools; • Applied computing → Education;

Keywords

Psychophysiology, Signal Processing, ECG, HRV, Human Factors, Wearables, Sensors

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1 Benefits

Human-computer interaction is leveraging psychophysiological measures both for studying user experience[9, 12] and as implicit

interaction mechanisms[3]; yet, many individuals usually learn electrophysiological signal processing without understanding the psychological and physiological bases and foundations of psychophysiology. While the potential benefits of incorporating psychophysiological measures in human-computer interaction are considerable, applying them effectively requires careful understanding of measurement and interpretation of psychophysiology signals[2]. Participants will learn about key psychophysiological principles and be guided through hands-on activities to record, process, and interpret their own signals. Commonly measured signals include heart rate (HR), accelerometry, respiration, brain waves (EEG/ERP), skin conductance (GSR/SCL), muscle contraction (EMG), eye-tracking, and sleep (PSG). This year's hands-on activities showcase electrocardiography (ECG) and heart rate variability (HRV). The course will focus on applied psychophysiology, signal recording, and signal processing as well as experimental designs and applications.

2 Intended Audience(s)

This cross-disciplinary course is intended for individuals interested in any of the following domains: wearables, biosensing, quantified self, brain-computer interfaces, biofeedback, health and medical technology (medtech), user experience, interactive applications, embedded sensors, augmented/virtual environments, human factors, cognitive ergonomics, internet of things, ubiquitous multimedia, electronic textiles, interaction design and computation arts, and tangible computing. This course is best suited for curious participants with inquisitive minds who like to roll up their sleeves and get their hands dirty.

3 Prerequisites

This introductory course assumes no prior knowledge. Attendees with previous experience in the field of psychophysiology are encouraged to attend to update their knowledge about recent advances in wearable technology.

4 Content

This course covers the following topics:

Applied Psychophysiology

- Overview of the autonomic nervous system
- Electricity and physiological signals
- Sensors for electrophysiological measurement
- Quantification of behavior

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Signal Recording

- Key principles of psychophysiology
- Sampling matters
- Rest, arousal, and active states
- Isolating and eliciting the phenomenon of interest
- Hands-on Wearables Laboratory: See your physiology

Signal Processing

- Cleaning noise and analyzing signals
- Same signal, different indices
- Scoring and data reduction
- Considerations for adequate trials & sampling time
- Hands-on Wearables Laboratory: Score your physiology

Experiments and Applications

- Translating and interpreting signals
- Experimental paradigms into the wild
- Quantifying change (Law of Initial Values)
- Individual vs. group differences and potential for bias
- Hands-on Wearables Laboratory: Interpret your physiology

5 Practical work

The Hands-on Wearables Laboratory component that is integrated throughout the course features engaging, sequential activities where participants will gain first-hand experience using software and equipment to measure their own physiological signals. Signals used to capture heart rate will be highlighted at CHI 2025. Participants will have an opportunity to experience psychophysiological phenomena, with demonstrations of the Diving Reflex and Baroreceptor Response. Learning activities showcase how to ensure good quality signals, how to prepare signals for analysis, how to interpret the signals, and how to integrate information into human-computer interaction systems.

6 Instructor background

Jennifer J. McGrath, PhD MPH, the inaugural PERFORM Chair in Childhood Preventive Health and Data Science, is a Professor in the Departments of Psychology at Concordia University, Montreal, Quebec and Binghamton University SUNY, New York, USA. She has taught Psychophysiology and advanced methodological workshops for the past 25 years. With extensive experience using wearables to measure heart rate variability, blood pressure, and polysomnography in children and adolescents, she has pioneered rigorous pediatric ambulatory measurement standards [6, 7, 10, 13]. Her research broadly focuses on the pathogenesis of subclinical cardiovascular disease markers that emerge during childhood and adolescence.

Marc N. Jarczok, PhD, is a full time researcher in the department of Psychosomatic Medicine and Psychotherapy at the University Medical Center Ulm, Germany. He has over a decade of experience in teaching and conducting research in the field of psychophysiology, focusing on the relation between workplace stress and individuals' health. Dr. Jarczok is an expert in using advanced statistical methods and wearable technologies, including heart rate variability and blood pressure monitoring, to investigate health outcomes in diverse populations [5]. His research primarily explores how psychosocial stress impacts physiological processes, particularly the

early onset of cardiovascular and immune regulatory disorders. Dr. Jarczok's work is pivotal in identifying early physiological markers of disease, to improve workplace interventions and promote long-term health outcomes across various occupational groups.

Paula Lago, PhD, is an Assistant Professor in the Department of Electrical and Computer Engineering at Concordia University. Her research is focused on understanding and measuring the relation between behaviors and health outcomes in ecological and cost-effective ways by using wearables, sensors at home, and machine learning [1, 4, 8, 11]. She was a volunteer on the SIGCHI Communications Team from 2018 to 2024.

7 Resources

Course materials and additional details (slides, worksheets, software) will be available at www.wearableslaboratory.org. Dr. McGrath is Director of the Pediatric Public Health Psychology Lab www.pphplab.org and Data Scientifique www.datascientifique.ca. Drs. Jarczok <https://www.uniklinik-ulm.de/psychosomatische-medizin-und-psychotherapie/ag-klinisch-experimentelle-stressforschung.html> and Lago www.paulalago.co share data, code, and other relevant resources on their websites.

8 Accessibility

Instructors will strive to implement universal design principles into course activities and make shared course materials adhere to accessibility requirements.

Acknowledgments

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